Speech Acoustics to rt-MRI Articulatory Dynamics Inversion with Video Diffusion Model

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Motivation

Acoustics-to-Articulatory Dynamics
Generating articulatory dynamics from speech signals and static vocal tract anatomy.

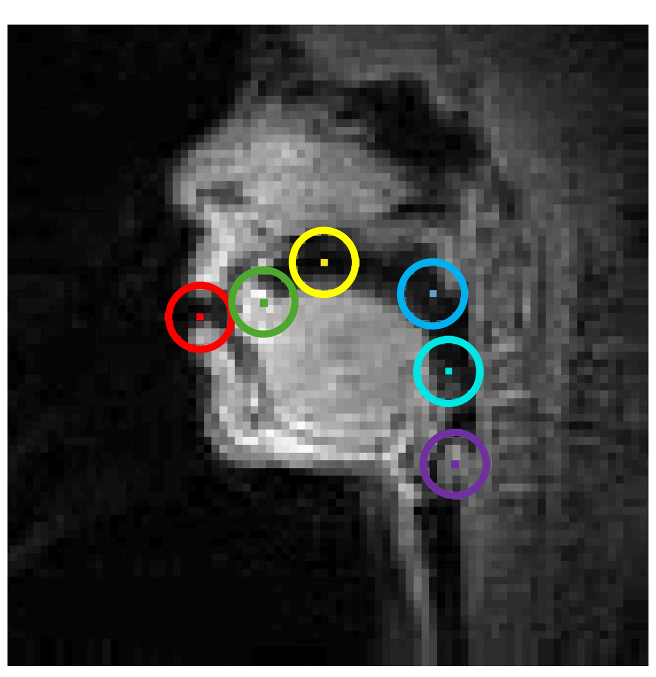
Applications
- Speech sound disorder intervention
- Second language acquisition
- Digital avator making

Chllenges
- The acoustic and articulatory dynamics do not necessarily have a strictly one-to-one mapping.
- Poor generalization to unseen speech or speakers
- Lack of accurate and objective evaluation metrics designed for articulatory dynamics

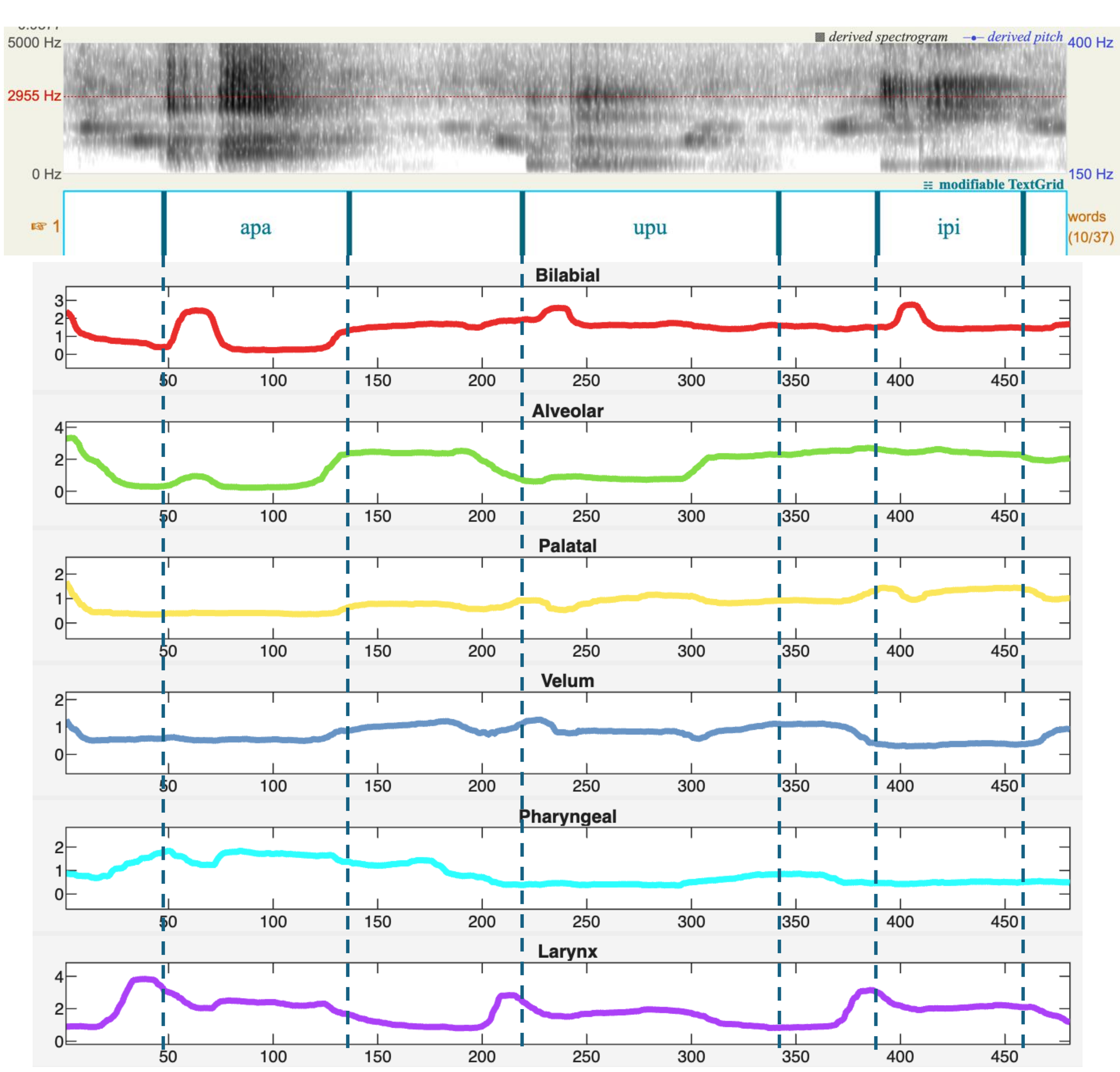
Experiments

Datasets – 75 Speakers Dataset[1]
- 75 speakers from diverse linguistic background
- Containing speech, mid-sagittal vocal tract rt-MRI, and 3D static MRI
- Speech: running speech (sentence & passage reading, spontaneous speech), consonant-vowel sequences

Eveluation – ROIs Pixel Intensity



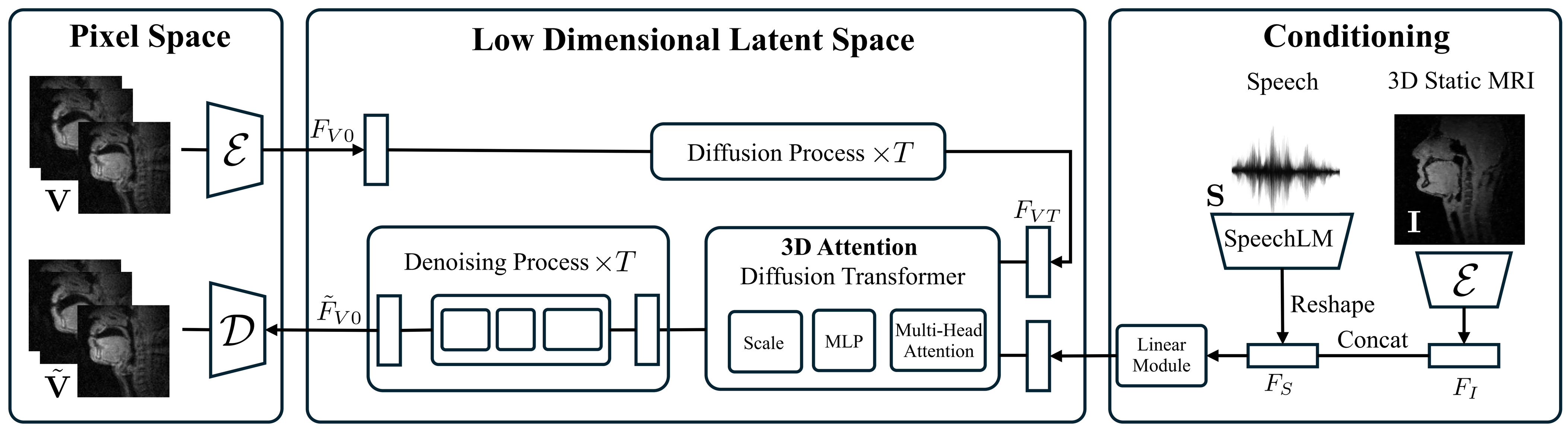
- Six regions of interest (ROIs)
- Pixel intensity through rt-MRI sequence
- Manual annotation from three annotators trained in phonetics and speech scienc



[1] Lim, Yongwan, et al. "A multispeaker dataset of raw and reconstructed speech production real-time MRI video and 3D volumetric images." *Scientific data* 8.1 (2021): 187.
[2] Reda, Fitsum, et al. "Film: Frame interpolation for large motion." *European Conference on Computer Vision*. Cham: Springer Nature Switzerland, 2022.

Methodology

Modeling – Conditional Video Diffusion Model



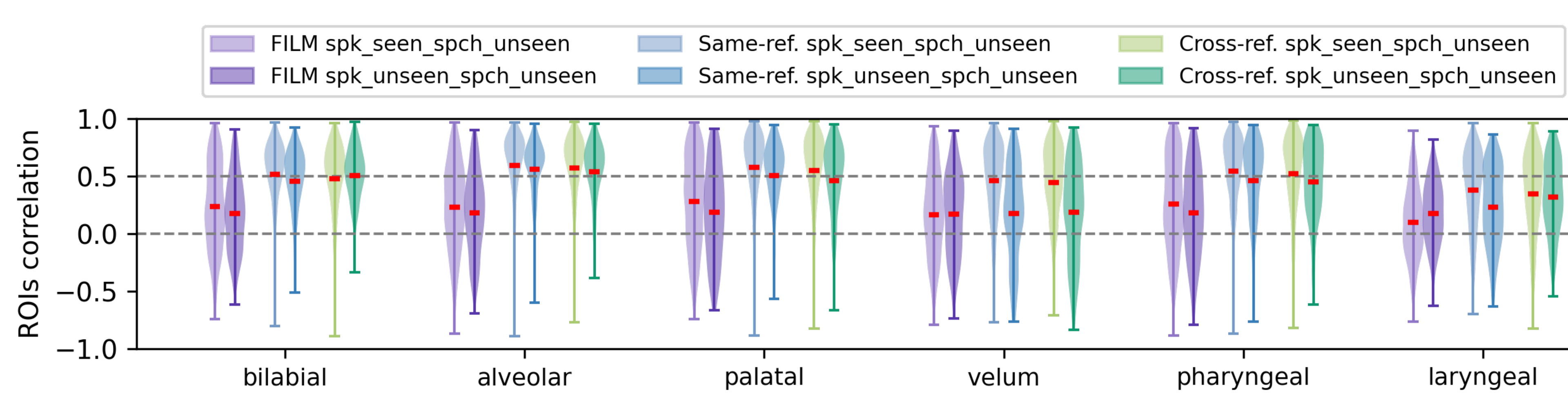
Time-aligned data point:
Speech waveform $\mathbf{S} \in \mathbb{R}^{t_S}$, 3D static MRI $\mathbf{I} \in \mathbb{R}^{h_I \times w_I \times c}$,
Rt-MRI sequence $\mathbf{V} \in \mathbb{R}^{t_V \times h_V \times w_V \times c}$
Denoising Process: $p_\theta(F_{Vt-1}|F_{Vt}, F_S, F_I) = \mathcal{N}(F_{Vt-1}; \mu_\theta(F_{Vt}, t, F_S, F_I), \beta_t I)$
Objective Function:

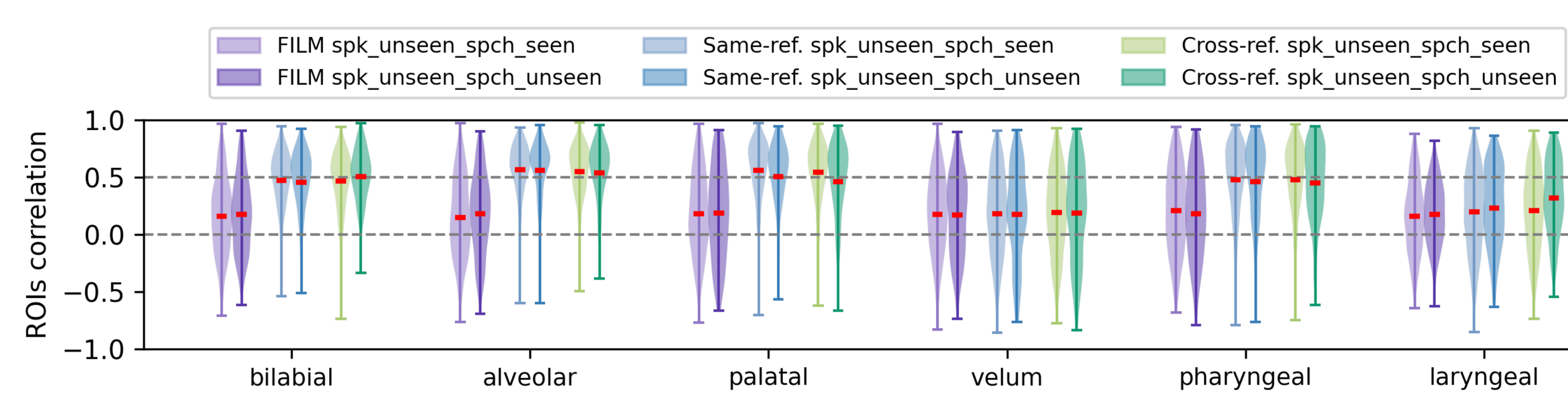
$$L_{VLB} = -\log p_\theta(F_{V0}|F_{V1}) + D_{KL}(q(F_{Vt-1}|F_{Vt}, F_{V0}) \,||\, p_\theta(F_{Vt-1}|F_{Vt}))$$

$$L_{MSE} = \mathbb{E}_{F_{V0}, \epsilon \sim \mathcal{N}(0,1), t}\left[\|\epsilon - \epsilon_\theta(F_{Vt}, t, F_S, F_I)\|_2^2\right]$$

Experimental Settings
- Speaker vs. Speech, Seen vs. Unseen: we splite train/val/test sets and preserve unseen speaker and unseen speech to evaluate model's generalize capability.
- Same vs. Cross –reference: we generate the rt-MRI video with the input of speech and the reference image taken from the rt-MRI instead of the 3D static MRI, aiming to assess model's theoretical performance upper bound.

Baseline: FILM[2] – video interpolation





Findings
- Both same-ref. and cross-ref. systems show superior pixel intensity correlation within ROIs than the FILM in the pharyngeal region and front articulators: bilabial, alveolar, and palatal region;
- In the "unseen speaker" condition, the velar and laryngeal regions demonstrate significant correlation degradation. Potential linguistic reason: 1) constrictions formed by the tongue dorsum against the velum are known to vary substantially in location as a function of the flanking vowels, 2) no English phonemes actively target mid-sagittal constrictions in the laryngeal region;
- Under the "unseen speech" setting, the video generation model maintains decent pixel intensity correlations within ROIs, indicating that our model has the capability to generate reasonable rt-RMI under the low speech resource setting.

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